

01

Start here: your last hour

Scope. A corrected revision guide based on all 11 supplied note files: curves, fields, differential calculus, vector calculus, surfaces, multiple/line/surface integrals, numerical series, power series, and Fourier series. Small supporting additions are labelled. This is a revision aid for that scope, not a prediction of exam questions.

Use the time you have left

Minutes	What to do
0–5	Read the corrections below; recall each formula before looking.
5–17	Pages 2–4: differentiability, Hessian, parametrizations, normals.
17–34	Pages 5–7: Jacobians, conservative fields, Green/Gauss/Stokes.
34–47	Pages 8–9: choose a series test; solve both power-series endpoints.
47–56	Page 10: Fourier coefficients, parity, jump values, Parseval.
56–60	Close the guide. Reproduce the five formulas below and their conditions.

Important corrections to your notes

- Series:** $\sum a_n$ convergent $\Rightarrow a_n \rightarrow 0$. The reverse is false: $\sum 1/n$ diverges.
- Change of variables:** include $|\det D\Phi|$ in every multiple integral.
- Schwarz's theorem:** continuous *second* partial derivatives near the point suffice for equality of mixed partials. $f \in C^0$ alone does not.
- Differentiability:** the remainder is $o(\|h\|)$; the second-order Taylor Hessian is evaluated at the *expansion point*.
- Conservative does not mean radial.** It means $F = \nabla U$. Curl-free implies conservative on an open simply connected domain, with $F \in C^1$.
- Root test:** use $\sqrt[n]{|a_n|}$, not $\sqrt{|a_n|}$. Ratio/root limit 1 is inconclusive; the limit of the ratio is not the sum.
- Monotone sequences:** an increasing sequence has a finite limit or tends to $+\infty$, not $-\infty$.
- Orientations and hypotheses:** Gauss needs a closed boundary with outward normal; Stokes needs compatible boundary orientation; fields must be regular on a neighborhood of the entire region/surface used.

Five formulas to know cold

$$\begin{aligned}\int_{\gamma} f \, ds &= \int_a^b f(\gamma(t)) \|\gamma'(t)\| \, dt, \\ \int_{\gamma} F \cdot dr &= \int_a^b F(\gamma(t)) \cdot \gamma'(t) \, dt, \\ \iint_D f \, dA &= \iint_E f(\Phi(u, v)) |\det D\Phi| \, du \, dv, \\ \iint_{\Sigma} f \, dS &= \iint_E f(r(u, v)) \|r_u \times r_v\| \, du \, dv, \\ \iint_{\Sigma} F \cdot n \, dS &= \iint_E F(r(u, v)) \cdot (r_u \times r_v) \, du \, dv.\end{aligned}$$

Memory rule: scalar integrals use a *norm*; work and flux use an *oriented vector*. The last formula assumes the cross product gives the requested orientation.

02

Limits and differential calculus

Fields, continuity, and limits

A scalar field is $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$; a vector field is $F : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$. A level set is $\{x : f(x) = c\}$. A radial scalar field has form $f(x) = g(\|x\|)$.

f is continuous at p if $\lim_{x \rightarrow p} f(x) = f(p)$. A multivariable limit must be the same along *every* approach in its domain.

- To **disprove** a limit: try axes, lines $y = mx$, then curves such as $y = x^2$.
- To **prove** a limit L : bound $|f(p+h) - L| \leq g(\|h\|)$ with $g(r) \rightarrow 0$.
- At the origin use $x = r \cos \theta$, $y = r \sin \theta$. A bound independent of θ proves the limit. Checking each fixed angle alone is insufficient.

Example. $f(x, y) = xy/(x^2 + y^2)$ has no limit at 0: on $y = 0$ it is 0, on $y = x$ it is $1/2$. Also $x^2y/(x^4 + y^2)$ tends to 0 on every straight line through 0, but equals $1/2$ on $y = x^2$.

Partial derivatives, directional derivatives, differentiability

$$f_{x_i}(p) = \lim_{t \rightarrow 0} \frac{f(p + te_i) - f(p)}{t}, \quad D_v f(p) = \lim_{t \rightarrow 0} \frac{f(p + tv) - f(p)}{t}.$$

The definition of differentiability is

$$\boxed{f(p+h) = f(p) + \nabla f(p) \cdot h + o(\|h\|)}.$$

Test at a suspect point: compute the partials *from the definition*, then test

$$\frac{f(p+h) - f(p) - \nabla f(p) \cdot h}{\|h\|} \rightarrow 0.$$

C^1 near $p \Rightarrow$ differentiable at $p \Rightarrow$ continuous at p . Existence of all partial derivatives, or even all directional derivatives, does not suffice.

If f is differentiable, $D_v f(p) = \nabla f(p) \cdot v$. For *unit* v , its largest value is $\|\nabla f(p)\|$, in the gradient direction when the gradient is nonzero.

Tangent planes and the chain rule

For a graph $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$,

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

A normal is $(f_x, f_y, -1)$. On a regular level surface $g(x, y, z) = c$, a normal is $\nabla g(p) \neq 0$, and the tangent plane is $\nabla g(p) \cdot (x - p) = 0$.

$$\frac{d}{dt} f(\gamma(t)) = \nabla f(\gamma(t)) \cdot \gamma'(t), \quad D(G \circ F)(p) = DG(F(p)) DF(p).$$

Here $DF = (\partial F_i / \partial x_j)$ has m rows and n columns for $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Second-order Taylor expansion and Schwarz

For $f \in C^2$ near p , mixed partials agree and the Hessian $H_f = (f_{x_i x_j})$ is symmetric:

$$f(p+h) = f(p) + \nabla f(p) \cdot h + \frac{1}{2} h^T H_f(p) h + o(\|h\|^2).$$

In two variables the quadratic term is $\frac{1}{2}(f_{xx}h_1^2 + 2f_{xy}h_1h_2 + f_{yy}h_2^2)$, with all derivatives at p .

03

Extrema and vector differential operators

Stationary points: the complete routine

1. Determine the domain and where differentiability fails.
2. At interior differentiable points solve $\nabla f = 0$ (Fermat's necessary condition).
3. At each solution compute the Hessian; classify using the table below.
4. For *absolute* extrema, also inspect boundaries, corners, singular points, and any relevant behavior at infinity. Compare function values.

For $f \in C^2$ at a stationary point p , set $\Delta = f_{xx}f_{yy} - f_{xy}^2$, evaluated at p .

Condition	Conclusion
$\Delta > 0, f_{xx} > 0$	Strict local minimum
$\Delta > 0, f_{xx} < 0$	Strict local maximum
$\Delta < 0$	Saddle point
$\Delta = 0$	Inconclusive: inspect $f(p+h) - f(p)$ directly

In n dimensions: positive definite Hessian $\Rightarrow$ strict minimum; negative definite $\Rightarrow$ strict maximum; eigenvalues of both signs $\Rightarrow$ saddle. A semidefinite Hessian is inconclusive. Opposite signs still prove a saddle even if another eigenvalue is zero.

Example. At 0, $x^4 + y^4$ has a strict minimum, $-x^4 - y^4$ a strict maximum, and $x^4 - y^4$ a saddle. All have zero Hessian there.

Why the Hessian test works: at a stationary point the linear Taylor term vanishes. A definite quadratic term dominates the $o(\|h\|^2)$ remainder.

Useful additions: boundary extrema and constraints

Weierstrass: a continuous real function on a nonempty compact set attains its absolute minimum and maximum. Parametrize each boundary piece and solve a one-variable problem, including its endpoints.

Lagrange multipliers: on $g(x) = c$, a constrained extremum with $\nabla g \neq 0$ satisfies

$$\nabla f = \lambda \nabla g, \quad g = c.$$

These are candidates, not automatically extrema. Inspect points with $\nabla g = 0$ separately. For several regular constraints use $\nabla f = \sum_j \lambda_j \nabla g_j$, requiring independent constraint gradients.

Example. For $f(x, y) = xy$ on $x^2 + y^2 = 1$, solve $(y, x) = 2\lambda(x, y)$. The candidates have $y = \pm x$, giving maximum $1/2$ and minimum $-1/2$.

Divergence, curl, and identities

For $F = (P, Q, R)$,

$$\operatorname{div} F = P_x + Q_y + R_z, \quad \operatorname{curl} F = (R_y - Q_z, P_z - R_x, Q_x - P_y).$$

In the plane, $\operatorname{div}(P, Q) = P_x + Q_y$ and scalar $\operatorname{curl}(P, Q) = Q_x - P_y$. Positive planar curl corresponds to local counterclockwise circulation. In 3D curl is a *vector*, not a positive/negative scalar.

$$\begin{aligned} \operatorname{curl}(\nabla f) &= 0, & \operatorname{div}(\operatorname{curl} F) &= 0, \\ \operatorname{curl}(fF) &= \nabla f \times F + f \operatorname{curl} F, & \operatorname{div}(fF) &= \nabla f \cdot F + f \operatorname{div} F. \end{aligned}$$

For the first two identities assume the relevant functions are C^2 ; the product identities need C^1 . The Laplacian is $\Delta f = \operatorname{div} \nabla f = \sum_i f_{x_i x_i}$.

04

Curves and surface geometry

Curves: definitions and length

A parametrized curve is $\gamma : [a, b] \rightarrow \mathbb{R}^n$; its trace is $\gamma([a, b])$. It is closed if $\gamma(a) = \gamma(b)$, simple if there are no repeated points except a possible common endpoint, and regular if $\gamma \in C^1$ with $\gamma'(t) \neq 0$. Piecewise regular curves are handled piece by piece.

$$\tau(t) = \frac{\gamma'(t)}{\|\gamma'(t)\|}, \quad ds = \|\gamma'(t)\| dt, \quad L(\gamma) = \int_a^b \|\gamma'(t)\| dt.$$

The tangent line at t_0 is $\gamma(t_0) + s\gamma'(t_0)$.

Curve	Parametrization / useful fact
Segment A to B	$\gamma(t) = A + t(B - A)$, $0 \leq t \leq 1$; length $\ B - A\ $.
Circle, radius a	$(x_0 + a \cos t, y_0 + a \sin t)$, $0 \leq t \leq 2\pi$: counterclockwise.
Ellipse	$(a \cos t, b \sin t)$; speed $\sqrt{a^2 \sin^2 t + b^2 \cos^2 t}$.
Graph $y = f(x)$	$(t, f(t))$; length $\int_a^b \sqrt{1 + f'(t)^2} dt$.
Helix	$(a \cos t, a \sin t, bt)$; constant speed $\sqrt{a^2 + b^2}$.
Spirals	$(t \cos t, t \sin t)$ has speed $\sqrt{1 + t^2}$; $(e^t \cos t, e^t \sin t)$ has speed $\sqrt{2}e^t$.

A regular one-to-one change of parameter preserves length. Reversing the parameter reverses orientation; tracing a curve twice counts it twice in an integral.

Surface patches and orientation

A regular surface patch $r : E \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is C^1 and has

$$N = r_u \times r_v \neq 0, \quad n = \frac{N}{\|N\|}, \quad dS = \|N\| du dv.$$

Use patches that cover the surface once (apart from harmless seams/boundaries). A surface is orientable if it admits a continuous choice of unit normal. Choosing one of the two directions orients it. A Möbius strip is not orientable.

At $p = r(u_0, v_0)$ the tangent plane is $N(u_0, v_0) \cdot (x - p) = 0$. Reversing the cross-product order changes the normal's sign.

Surface	Parametrization and vector area element
Graph $z = g(x, y)$	$r(x, y) = (x, y, g)$; upward $N = (-g_x, -g_y, 1)$; $dS = \sqrt{1 + g_x^2 + g_y^2} dx dy$.
Cylinder $x^2 + y^2 = a^2$	$r(\theta, z) = (a \cos \theta, a \sin \theta, z)$; $r_\theta \times r_z = (a \cos \theta, a \sin \theta, 0)$ is outward; $dS = a d\theta dz$.
Sphere of radius a	$r(\phi, \theta) = (a \sin \phi \cos \theta, a \sin \phi \sin \theta, a \cos \phi)$; $r_\phi \times r_\theta$ is outward; $dS = a^2 \sin \phi d\phi d\theta$.

For the sphere, $0 \leq \phi \leq \pi$, $0 \leq \theta < 2\pi$; the standard coordinates degenerate at the poles, which are handled by other patches or ignored as area-zero sets when integrating.

Watch out. A sphere's *surface* element is $a^2 \sin \phi d\phi d\theta$; a spherical *volume* element also includes the radial differential.

05

Double and triple integrals

Draw the region before choosing bounds

If $D = \{(x, y) : a \leq x \leq b, \alpha(x) \leq y \leq \beta(x)\}$, then

$$\iint_D f \, dA = \int_a^b \int_{\alpha(x)}^{\beta(x)} f(x, y) \, dy \, dx.$$

For the opposite order describe D with y outermost and x between functions of y . Split the region where the bounding curves change. The inner bounds may depend on outer variables, never on the variable already being integrated out.

For continuous functions on compact ordinary regions these iterated formulas apply. More generally, Fubini applies to absolutely integrable functions; Tonelli applies to nonnegative functions, allowing infinite values.

Example. The triangle $0 \leq y \leq x \leq 1$ gives $\int_0^1 \int_0^x f \, dy \, dx = \int_0^1 \int_y^1 f \, dx \, dy$.

Change of variables: transform both the function and the region

For a C^1 one-to-one map $\Phi : E \rightarrow D$ with nonzero determinant (up to the usual measure-zero seams),

$$\int_D f(x) \, dx = \int_E f(\Phi(u)) |\det D\Phi(u)| \, du.$$

This applies in both two and three dimensions. The determinant is always in *absolute value* for volume/area substitutions.

Coordinates	Substitution	Jacobian
Polar	$x = x_0 + r \cos \theta, y = y_0 + r \sin \theta$	r
Elliptic scaling	$x = x_0 + ar \cos \theta, y = y_0 + br \sin \theta$	$ ab r$
Cylindrical	$(x, y, z) = (r \cos \theta, r \sin \theta, z)$	r
Spherical	$(x, y, z) = (\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi)$	$\rho^2 \sin \phi$

Here $r, \rho \geq 0$, θ is azimuth, and $0 \leq \phi \leq \pi$ is the angle from the positive z axis. Full circles use an angular interval of length 2π .

Triple integrals: columns or slices

For columns, $(x, y) \in D$ and $\alpha(x, y) \leq z \leq \beta(x, y)$; for slices, $c \leq z \leq d$ and $(x, y) \in D_z$:

$$\iiint_{\Omega} f \, dV = \iint_D \int_{\alpha}^{\beta} f \, dz \, dA, \quad \iiint_{\Omega} f \, dV = \int_c^d \iint_{D_z} f \, dA \, dz.$$

Example. For $\Omega = \{x^2 + y^2 \leq z \leq 4\}$, cylindrical bounds are $0 \leq r \leq 2, 0 \leq \theta \leq 2\pi, r^2 \leq z \leq 4$. Thus

$$\text{Vol}(\Omega) = \int_0^{2\pi} \int_0^2 \int_{r^2}^4 r \, dz \, dr \, d\theta = 8\pi.$$

Example. On the disk $x^2 + y^2 \leq a^2$, $\iint_D (x^2 + y^2) \, dA = \int_0^{2\pi} \int_0^a r^3 \, dr \, d\theta = \pi a^4/2$.

Fast checks and useful additions

Area is $\iint_D 1 \, dA$; volume is $\iiint_{\Omega} 1 \, dV$. An integrable odd function has integral zero on a region invariant under the corresponding reflection. For density δ , mass $M = \int \delta$, and center coordinate $\bar{x} = M^{-1} \int x\delta$ (similarly for y, z).

Improper radial integrals: in dimension d , $\|x\|^{-p}$ is integrable near 0 if $p < d$, and at infinity if $p > d$, by comparing $\int r^{d-1-p} \, dr$.

06

Line integrals, potentials, and Green

Type 1 versus type 2

$$\int_{\gamma} f \, ds = \int_a^b f(\gamma(t)) \|\gamma'(t)\| \, dt, \quad \int_{\gamma} F \cdot dr = \int_a^b F(\gamma(t)) \cdot \gamma'(t) \, dt.$$

Type 1 measures quantities such as wire mass and is unchanged by reversing orientation. Type 2 is work/circulation and changes sign. In the plane, $F \cdot dr = P \, dx + Q \, dy$; in 3D add $R \, dz$.

Conservative fields and path independence

On an open path-connected domain, F is conservative if $F = \nabla U$ for a potential U . Then

$$\boxed{\int_A^B F \cdot dr = U(B) - U(A)}, \quad \oint_{\gamma} F \cdot dr = 0.$$

For continuous fields, existence of a potential, path independence, and vanishing integrals on every closed piecewise smooth loop are equivalent. Potentials differ by a constant on a connected domain.

For $F \in C^1$, conservative $\Rightarrow \operatorname{curl} F = 0$. Conversely, on an *open simply connected domain*, $\operatorname{curl} F = 0 \Rightarrow$ conservative (in 2D/3D). Simply connected means every closed loop contracts to a point inside the domain.

Watch out. Check the *domain*, including excluded points/axes, before using the converse. A field need not be radial to be conservative. Radial fields $g(\|x\|)x/\|x\|$ are a useful special case, with potential $G(\|x\|)$ when $G' = g$.

How to find a potential

In 2D integrate $U_x = P$: $U(x, y) = \int P(x, y) \, dx + h(y)$. Differentiate in y , set $U_y = Q$, and determine h . In 3D start with an unknown function $h(y, z)$ and match the other two components. Finally check *all* derivatives.

Example. $F = (2xy, x^2 + 2y)$ has $Q_x - P_y = 2x - 2x = 0$ on $\mathbb{R}^2$. Integrating gives $U = x^2y + y^2 + C$. Work from $(0, 0)$ to $(1, 2)$ is 6.

Green's theorem: turn a closed planar integral into area

Let D be bounded with piecewise smooth boundary and $P, Q \in C^1$ on a neighborhood of $\overline{D}$. With D on the *left* as the boundary is traversed,

$$\boxed{\oint_{\partial D} P \, dx + Q \, dy = \iint_D (Q_x - P_y) \, dA.}$$

Outer boundary: counterclockwise. Hole boundaries: clockwise. Clockwise traversal of an entire simple outer curve changes the sign.

$$\operatorname{Area}(D) = \frac{1}{2} \oint_{\partial D} (x \, dy - y \, dx) = \oint_{\partial D} x \, dy.$$

The planar flux form is $\oint_{\partial D} P \, dy - Q \, dx = \iint_D (P_x + Q_y) \, dA$, for the same positive traversal and outward normal.

Example. $F = (-y, x)$ around the circle of radius a counterclockwise gives $\oint F \cdot dr = \iint_D 2 \, dA = 2\pi a^2$.

The standard hole counterexample

$$F(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right), \quad D = \mathbb{R}^2 \setminus \{0\}.$$

Curl is zero on D , but on $\gamma(t) = (\cos t, \sin t)$, $F(\gamma) \cdot \gamma' = 1$, so the circulation is 2π . Hence no global potential. Green cannot be applied across a disk containing the singularity.

07

Surface integrals, Gauss, and Stokes

Direct computation on a parametrized surface

For $r : E \rightarrow \Sigma$, $N = r_u \times r_v$,

$$\iint_{\Sigma} f \, dS = \iint_E f(r) \|N\| \, du \, dv, \quad \iint_{\Sigma} F \cdot n \, dS = \iint_E F(r) \cdot N \, du \, dv.$$

Choose the sign of N to match the requested normal. Surface area is the first formula with $f = 1$. Scalar surface integrals ignore orientation; flux reverses sign when the normal reverses.

For a graph $z = g(x, y)$ with *upward* normal,

$$\iint_{\Sigma} F \cdot n \, dS = \iint_D F(x, y, g(x, y)) \cdot (-g_x, -g_y, 1) \, dx \, dy.$$

Watch out. Do not normalize N and then forget $\|N\|$ from dS . The two factors cancel in the vector-area formula.

Divergence theorem (Gauss): closed-surface flux

For a bounded solid Ω with piecewise smooth boundary, $F \in C^1$ on a neighborhood of $\bar{\Omega}$, and *outward* normal,

$$\iint_{\partial\Omega} F \cdot n \, dS = \iiint_{\Omega} \operatorname{div} F \, dV.$$

If the surface is open, close it with convenient caps and subtract their *outward* fluxes. If there is a cavity, its outward normal for the solid points *into the cavity*. A singularity inside the volume invalidates direct application.

Example. For $F = (x, y, z)$ on the sphere of radius a , outward flux is $\int_{B_a} 3 \, dV = 4\pi a^3$. Directly, $F \cdot n = a$ and area is $4\pi a^2$.

Example. For $F = (0, 0, z)$ on the upper unit hemisphere with outward normal, close with the equatorial disk. Its outward normal is downward and its flux is 0 since $z = 0$. Thus hemisphere flux equals the half-ball volume, $2\pi/3$.

Stokes' theorem: boundary circulation equals curl flux

For an oriented piecewise smooth surface Σ and $F \in C^1$ on a neighborhood of it,

$$\oint_{\partial\Sigma} F \cdot dr = \iint_{\Sigma} (\operatorname{curl} F) \cdot n \, dS.$$

The boundary follows the right-hand rule: for an upward normal on a horizontal disk it is counterclockwise when viewed from above. All boundary components count. You may replace the spanning surface by an easier one with the same oriented boundary, provided the field is regular on that surface too.

Example. For $F = (-y/2, x/2, 0)$ and the unit circle counterclockwise viewed from above, use the disk: $\operatorname{curl} F = (0, 0, 1)$, so circulation is π .

Which method is fastest?

Given integral	First method to check
Work along an open path	Potential, otherwise direct parametrization
Circulation around a planar loop	Green, or a potential if one exists
Circulation around a space curve	Stokes with an easy spanning surface
Flux through a closed surface	Gauss if divergence is easy
Flux through an open surface	Direct graph/patch formula, or close and subtract caps
Integral of curl through a surface	Stokes if its boundary is simpler

Before using any theorem: name the region, check regularity/singularities, specify orientation, then substitute. Do not apply a flux theorem to a scalar surface integral.

08

Sequences and numerical series

Sequences and the first series test

A nondecreasing sequence has limit $\sup_n a_n$ in $\mathbb{R} \cup \{+\infty\}$; if bounded above it converges in $\mathbb{R}$. A nonincreasing sequence is analogous with infimum and $-\infty$. Every convergent sequence is bounded; bounded does not imply convergent.

A series converges when its partial sums $S_N = \sum_{n=n_0}^N a_n$ have a finite limit.

$$\sum a_n \text{ convergent} \Rightarrow a_n \rightarrow 0. \quad \text{The converse is false.}$$

For nonnegative terms, partial sums increase: the series either has a finite sum or tends to $+\infty$. A signed divergent series may instead oscillate.

Benchmarks and positive-term tests

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q} \quad (|q| < 1), \quad \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges iff } p > 1.$$

- **Comparison:** eventually $0 \leq a_n \leq b_n$ and $\sum b_n < \infty \Rightarrow \sum a_n < \infty$. If $a_n \geq b_n \geq 0$ and $\sum b_n = +\infty$, then $\sum a_n = +\infty$.
- **Limit comparison:** for eventually positive a_n, b_n , if $a_n/b_n \rightarrow c \in (0, \infty)$, the two series have the same convergence behavior. Asymptotic equivalence is the case $c = 1$.
- **Integral test (useful addition):** for positive decreasing continuous f , $\sum f(n)$ and $\int^{\infty} f(x) dx$ converge together. In particular $\sum_{n \geq 2} 1/[n(\log n)^p]$ converges iff $p > 1$.

For arbitrary signs apply ratio/root tests to *absolute values*:

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \quad \text{or} \quad L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

If $L < 1$, absolute convergence; if $L > 1$ (including infinity), divergence because terms do not tend to zero; if $L = 1$, no conclusion. The ratio requires eventually nonzero denominators. The root test also works with limsup: less than 1 gives absolute convergence, greater than 1 gives divergence.

Alternating series and absolute convergence

Absolute convergence, $\sum |a_n| < \infty$, implies convergence. Conditional convergence means convergence with $\sum |a_n| = \infty$.

Leibniz: if $b_n \geq 0$ is eventually nonincreasing and $b_n \rightarrow 0$, then $\sum (-1)^n b_n$ converges. Once in the decreasing range, its truncation error satisfies

$$|S - S_N| \leq b_{N+1},$$

and has the sign of the first omitted term. Alternating signs alone are not enough.

Choose a test from the shape of the term

Term structure	Try first
Rational expression in n	Dominant powers / comparison with $1/n^p$
Factorials or exponentials	Ratio test
An expression raised to n	Root test
$(-1)^n$ times a positive term	Absolute convergence, then Leibniz
Logarithms with ratio limit 1	Comparison or integral test

Example. $n/(n^3 + 1) \sim 1/n^2$: converges. For $n!/n^n$, the ratio is $(n/(n+1))^n \rightarrow e^{-1}$: converges. $\sum (-1)^{n-1}/n$ converges conditionally, with error $\leq 1/(N+1)$.

Why the term test is necessary: $a_n = S_n - S_{n-1} \rightarrow S - S = 0$. This implication cannot be reversed.

09

Power series and analyticity

Radius and interval of convergence

For $\sum_{n=0}^{\infty} a_n(x - x_0)^n$,

$$R = \frac{1}{\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}}.$$

Use $1/0 = \infty$ and $1/\infty = 0$. If the coefficient ratio exists with eventually nonzero coefficients,

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \Rightarrow R = 1/L.$$

- $|x - x_0| < R$: absolute convergence.
- $|x - x_0| > R$: divergence.
- $x = x_0 \pm R$ for $0 < R < \infty$: substitute separately and use numerical-series tests.

Routine: identify x_0 and a_n ; compute R ; test the two endpoints; report the interval with the correct open/closed brackets. If $R = 0$, only the center converges. If $R = \infty$, every real x converges.

Example. For $\sum_{n=1}^{\infty} (x-2)^n/(n3^n)$, $R = 3$. At $x = 5$ it is $\sum 1/n$, divergent; at $x = -1$ it is $\sum (-1)^n/n$, convergent. Interval: $[-1, 5)$.

Operations inside the radius

A power series converges uniformly and absolutely on every closed interval $|x - x_0| \leq r < R$. Its sum is continuous and can be differentiated/integrated term by term there:

$$f'(x) = \sum_{n=1}^{\infty} n a_n (x - x_0)^{n-1},$$

$$\int_{x_0}^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} (x - x_0)^{n+1}.$$

The differentiated/integrated series have the same radius, but may behave differently at endpoints. If two series with the same center have *different* radii, their sum has the smaller radius. With equal radii cancellation may increase it.

Watch out. Uniform convergence on every smaller closed interval does not assert uniform convergence on the full open convergence interval.

Standard expansions to recognize

$$\begin{aligned} \frac{1}{1-t} &= \sum_{n=0}^{\infty} t^n, & |t| < 1, \\ e^t &= \sum_{n=0}^{\infty} \frac{t^n}{n!}, & t \in \mathbb{R}, \\ \sin t &= \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)!}, & t \in \mathbb{R}, \\ \cos t &= \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n)!}, & t \in \mathbb{R}, \\ \log(1+t) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} t^n}{n}, & -1 < t \leq 1, \\ \arctan t &= \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{2n+1}, & -1 \leq t \leq 1. \end{aligned}$$

Useful derived formula: $\sum_{n=1}^{\infty} n t^{n-1} = 1/(1-t)^2$ for $|t| < 1$. Derive it by differentiating the geometric series.

Analytic functions

Analytic at x_0 means equal to a convergent power series in a neighborhood of x_0 ; necessarily $a_n = f^{(n)}(x_0)/n!$. Smoothness C^∞ alone does not imply analyticity.

A useful sufficient condition on a neighborhood is $|f^{(n)}(x)| \leq M A^n n!$ for every $n \geq 0$, with fixed $M, A > 0$. Taylor's remainder then tends to zero when $A|x - x_0| < 1$. The stronger bound $|f^{(n)}(x)| \leq C^n$ stated in your notes is sufficient too, but not necessary.

10

Fourier series: coefficients and convergence

Keep your notes' convention: the constant term is a_0 , not $a_0/2$. For a real T -periodic function, write $\omega = 2\pi/T$ and

$$f(x) \sim a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega x) + b_n \sin(n\omega x)].$$

Over any interval I of length T ,

$$a_0 = \frac{1}{T} \int_I f \, dx, \quad a_n = \frac{2}{T} \int_I f \cos(n\omega x) \, dx, \quad b_n = \frac{2}{T} \int_I f \sin(n\omega x) \, dx.$$

For $T = 2\pi$, use $\omega = 1$, $I = [-\pi, \pi]$, and factors $1/(2\pi), 1/\pi, 1/\pi$.

Parity, orthogonality, and the calculation routine

On a symmetric period, even f has $b_n = 0$; odd f has $a_0 = a_n = 0$. An even integrand can be integrated on $[0, T/2]$ and doubled. For $n, m \geq 1$,

$$\int_I \cos(n\omega x) \cos(m\omega x) \, dx = \frac{T}{2} \delta_{nm}, \quad \int_I \sin(n\omega x) \sin(m\omega x) \, dx = \frac{T}{2} \delta_{nm};$$

all sine-cosine products and integrals of single nonconstant modes vanish. This orthogonality produces the coefficient formulas by multiplication and integration.

1. Identify the period, periodic extension, and parity.
2. Compute the coefficients, splitting piecewise definitions and using integration by parts.
3. State the convergence value, including jumps and the periodic seam.
4. To evaluate a numerical sum, choose a useful x , or apply Parseval.

What kind of convergence do you have?

If the periodic function is piecewise C^1 with finite one-sided limits, then

$$S_N(x) \longrightarrow \frac{f(x^-) + f(x^+)}{2}.$$

At continuity points this is $f(x)$; at jumps it is the midpoint, regardless of the assigned value at the point. At $\pm T/2$ use the *periodic* one-sided limits. The Gibbs overshoot near jumps does not imply failure of pointwise convergence.

For $f \in L^2(I)$, $\|f - S_N\|_2 \rightarrow 0$, which alone does *not* imply pointwise or uniform convergence. Here $\|f\|_2^2 = \int_I |f|^2$ and $\|f\|_\infty = \sup_I |f|$ for bounded pointwise-defined f .

$$\int_I |f|^2 \, dx = Ta_0^2 + \frac{T}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \quad (\text{Parseval}).$$

For finite N , $\|f - S_N\|_2^2 = \|f\|_2^2 - Ta_0^2 - \frac{T}{2} \sum_{n=1}^N (a_n^2 + b_n^2) \geq 0$ (orthogonal projection / Bessel inequality). Riemann-Lebesgue says $a_n, b_n \rightarrow 0$ for $f \in L^1(I)$; coefficient decay alone does not prove pointwise convergence.

Two model examples

Sawtooth: the 2π -periodic extension of $f(x) = x$ on $(-\pi, \pi)$ is odd:

$$b_n = \frac{2}{\pi} \int_0^\pi x \sin(nx) \, dx = \frac{2(-1)^{n+1}}{n}, \quad x = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \sin(nx)}{n} \quad (-\pi < x < \pi).$$

At $\pm\pi$ the series converges to 0. Parseval gives $2\pi^3/3 = 4\pi \sum_{n \geq 1} 1/n^2$, hence $\sum 1/n^2 = \pi^2/6$.

Square wave: for $f = -1$ on $(-\pi, 0)$ and $f = 1$ on $(0, \pi)$, $a_0 = a_n = 0$ and $b_n = 2[1 - (-1)^n]/(\pi n)$. Thus only odd sine modes remain, with coefficient $4/(\pi n)$. At every jump the series converges to 0.